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Research Article

Identifying Lumbosacral Radiculopathies

An Optimal Electromyographic Screen

ABSTRACT

Dillingham TR, Lauder TD, Andary M, Kumar S, Pezzin LE, Stephens RT, Shannon S: Identifying lumbosacral radiculopathies: an optimal electromyographic screen. *Am J Phys Med Rehabil* 2000;79:496–503.

Objective: The objective of this study was to determine prospectively the optimal electromyographic screening examination of the lower limb that ensures identification of those lumbosacral radiculopathies that can be electrodiagnostically confirmed, yet minimizes the number of muscles studied.

Design: A prospective multicenter study was conducted from May 1996 to September 1997. Patients with suspected lumbosacral radiculopathy referred to participating electrodiagnostic laboratories were recruited and examined by needle electromyography using a standard set of muscles. Patients with electrodiagnostically confirmed lumbosacral radiculopathies were selected for analysis. Various muscle screens were tested against this group of patients with radiculopathies to determine the frequency with which each screen identified the patient with radiculopathy.

Results: There were 102 patients identified. When paraspinal muscles were one of the screening muscles, four-muscle screens identified 88–97% of the radiculopathies, five-muscle screens identified 94–98%, and six-muscle screens 98–100%. When paraspinal muscles were not part of the screen, identification rates were lower for all screens, and eight distal muscles were necessary to identify about 90% of the radiculopathies.

Conclusions: Six-muscle screens with paraspinal muscles yielded consistently high identification rates. Studying additional muscles produced no improvements in identification.

Key Words: Electrodiagnosis, Electromyography, Radiculopathy, Nerve Conduction, Lumbosacral Region, Radiculitis

The electrodiagnostic evaluation provides important information for the assessment of patients with lower-limb symptoms of pain, sensory loss, or weakness.[1–11] The differential diagnosis of these problems is quite large and includes lumbosacral radiculopathy, entrapment neuropathy, myopathy, neuromuscular junction disorder, motor neuron disease, polyneuropathy, musculoskeletal disorders, or central nervous system abnormalities such as myelopathy and stroke. The electrodiagnostician is compelled to perform a study sufficient either to identify confidently or to exclude lumbosacral radiculopathy. Studies that include a large number of muscles, however, are uncomfortable to the patient. For this reason, it is of great clinical interest to delineate an optimal electromyographic screening examination that allows the examiner to identify a lumbosacral radiculopathy (when one can be electrodiagnostically confirmed), yet minimize the number of muscles studied to prevent excessive patient discomfort.

The concept of a screening electromyographic (EMG) examination encompasses identification of a possible electrodiagnostically confirmable radiculopathy. If one of the muscles in the screen is abnormal, the screen must be expanded to exclude other diagnoses and to delineate fully the radiculopathy level. Some radiculopathies cannot be confirmed by needle EMG.[10, 11] Radiculopathies that exclusively cause sensory root involvement will not produce abnormal EMG findings.[10, 11] If the rate of denervation is balanced by reinnervation in the muscle, then spontaneous activity is less likely to occur. Although EMG findings correlate with spinal imaging (myelography and magnetic resonance imaging) and intraoperative findings, there are some radiculopathies that demonstrate denervation potentials on EMG but show no anatomic structural reasons to explain this muscle denervation.[12–14] This has been termed chemical or inflammatory radiculitis.[13, 14] Conversely, patients can have a clear structural cause of nerve root compromise and yet have a normal EMG.[12]

A screening EMG study determines whether the radiculopathy can be confirmed by EMG. If the radiculopathy cannot be confirmed, then presumably no amount of muscles examined by needle EMG can identify the radiculopathy. The process of identification, therefore, can be conceptualized as a conditional probability question: given that a lumbosacral radiculopathy can be confirmed by needle EMG, what is the minimum number of muscles that must be examined to recognize confidently this possibility?[15, 16]

Identifying a lumbosacral radiculopathy by EMG can have prognostic implications as well. Spengler and Freeman[17] and Spengler et al.[18] demonstrated in both retrospective and prospective studies that if a patient has a positive EMG, confirming the suspicion of lumbosacral radiculopathy, then the patient has a higher likelihood of a successful surgical outcome, particularly when this information is combined with imaging results and psychological testing. Tullberg et al.[19] showed that a negative electrodiagnostic study was significantly correlated with a poor surgical outcome. The EMG provides a physiologic assessment of root level motor axonal loss, and these findings complement other clinical and imaging parameters. The prognostic implications of a positive EMG further underscore the need to identify accurately and efficiently a lumbosacral radiculopathy when one can be confirmed.

Despite the great potential for minimizing physicians' time and patients' discomfort, the optimal number of muscles for evaluating patients with lower-limb symptoms has received little attention in the literature. Lauder et al.[16] retrospectively examined 247 patients with electrodiagnostically confirmed lumbosacral radiculopathies. Muscle screens with five muscles that included the lumbar paraspinal muscles yielded optimal identifications. Although this analysis provided new insights into how many muscles must be studied, the retrospective nature of the study implied inherent limitations.

The purpose of this investigation was to determine the optimal EMG screening examination of the lower limb that ensures detection of those lumbosacral radiculopathies that can be electrodiagnostically confirmed.

METHODS

Patient Selection

A prospective study was conducted between May 1996 and September 1997 at five medical centers treating diverse populations. Patients were asked to participate if they were referred for a suspected lumbosacral radiculopathy to the electrodiagnostic laboratories in the departments of Physical of Medicine and Rehabilitation at The Johns Hopkins University (Baltimore, MD), Ingham Regional Medical Center at Michigan State University (East Lansing, MI), Madigan Army Medical Center (Tacoma, WA), Womack Army Medical Center (Fort Bragg, NC), or Walter Reed Army Medical Center (Washington, DC). Subjects participated on a voluntary basis after giving written informed consent. The study protocol was approved by the appropriate Institutional Review committee at each of the participating institutions.

Patient Evaluation

Standardized procedures and data collection sheets were implemented at all participating centers. Patients completed a questionnaire that collected detailed information on symptoms and medical history. For all patients, a standardized physical ex-

amination was completed that consisted of the following: (1) a neurologic examination, including manual muscle testing, sensation (vibration and pinprick), and reflex assessments; (2) a musculoskeletal examination; and (3) special tests, such as straight-leg raise and Patrick's test. After the questionnaire and physical examination, electrodiagnostic evaluations, conducted by attending physicians or their supervised resident physicians, were performed for all patients. All participating attending physicians were certified by the American Board of Electrodiagnostic Medicine.

The standardized electrodiagnostic study consisted of at least (1) one lower-limb motor nerve conduction study, (2) one lower-limb sensory-nerve conduction study, and (3) needle EMG with either monopolar or concentric needles. Eleven muscles were part of the basic screening procedure and were studied for all subjects (whenever clinically possible). These muscles included the following: the lumbosacral paraspinal muscles (PSM), adductor longus (ADD), vastus medialis (VMED), vastus lateralis (VLAT), rectus femoris (RFEM), tensor fasciae latae (TFL), short head of the biceps femoris (SHBF), tibialis anterior (ATIB), tibialis posterior (PTIB), medial gastrocnemius (MGAS), and lateral gastrocnemius (LGAS). All data entry and coding, including scoring of abnormalities, were reviewed for accuracy and consistency. Information was recorded also regarding additional nerves or muscles studied at the electromyographer's discretion to ensure an accurate diagnosis.

Sample Selection

The sample used in these analyses consisted of all patients with an electrodiagnostically confirmed lumbosacral radiculopathy. Lumbosacral radiculopathies were defined by the presence of abnormalities in two or more muscles innervated by the same nerve root level but different peripheral nerves. Nonparaspinal lower-limb muscles were considered abnormal if they demonstrated the following characteristics: (1) positive sharp waves, (2) fibrillation potentials, (3) complex repetitive discharges, (4) high-amplitude, long-duration motor-unit action potentials, (5) increased polyphasic motor-unit action potentials (>30%), or (6) reduced neuropathic recruitment. Paraspinal muscles were considered abnormal if they showed fibrillation potentials, positive sharp waves, or complex repetitive discharges. Patients were also included in the study if only the PSM showed abnormalities. If the patient had a history of low-back surgery, paraspinal findings were ignored when determining whether a radiculopathy was electrodiagnostically identified. Increased insertional activity was not used for the inclusion criteria but was used in the identification analyses. For patients with bilateral lumbosacral radiculopathies, the side with the greatest number of muscles studied was chosen for analysis. Finally, subjects with demonstrated evidence of peripheral polyneuropathies, myopathies, or other generalized conditions were excluded from the study.

Individual Muscle and Muscle Screen Identifications

Individual muscle identification rates were determined by assessing the percentage of patients with an electrodiagnostically confirmed lumbosacral radiculopathy for whom the tested muscle was abnormal. For instance, if a patient with an L4 radiculopathy demonstrated spontaneous activity in the VMED, this patient would be considered successfully identified by any screen containing the VMED.

Muscle screens were created such that all potential radiculopathy levels (L3–S1) would be represented by their root level innervation patterns. If any muscle in the screen was abnormal for a particular patient, then the screen successfully identified that patient as having an electrodiagnostically confirmable lumbosacral radiculopathy.

Two types of muscle abnormalities were tested: (1) a more restrictive category consisting of only spontaneous activity (fibrillations and positive waves), and (2) a more inclusive neuropathic category reflecting other findings suggestive of radiculopathy (spontaneous activity, complex repetitive discharges, reduced recruitment, increased insertional activity, increased polyphasic potentials, or large-amplitude, long-duration motor-unit action potentials). Spontaneous activity can encompass other discharges such as myokymic discharges. Fibrillations and positive sharp waves are not unique to radiculopathy and are found in other conditions such as inflammatory myopathies. However, for the purposes of this study, spontaneous activity refers only to fibrillations and positive sharp waves that are reflective of axonal loss seen in radiculopathy. Quantitative EMG was not performed; rather, assessments regarding motor-unit morphology and recruitment were qualitatively derived by the experienced coinvestigators performing the study.

Sample-size calculations were performed a priori to fulfill several analytic goals of this prospective project. Based on the experience of previous studies,[15, 16] a sample size of 57 subjects was required to detect with 80% power the desired 15 percentage-point differences in identification rates between 3-muscle and 5-muscle screens for lumbosacral radiculopathies. The actual study sample size of 100 subjects, chosen primarily to meet the requirements of other analytic goals,[20, 21] yielded an estimated 94% power to detect the differences discussed above at the 5% critical level for unidirectional effects in a single sample.[22]

For data storage and statistical analyses, SPSS 7.5 for Windows (SPSS, Chicago, IL) was used. Statistical significance was considered at the $P < 0.05$ level for all statistical comparisons. Pearson's correlation

coefficients and χ^2 analyses were used for nonparametric statistical comparisons.

RESULTS

Subjects

There were 102 patients with electrodiagnostically confirmed lumbosacral radiculopathies. Males comprised 47% of the sample. The average age was 53 years (SD = 17 years) ranging from 23 to 87 years old. Ninety-six percent of all patients with lumbosacral radiculopathies had at least 1 muscle with spontaneous activity. Two or more muscles showed spontaneous activity in 75% of the sample. This indicates that the majority of patients with radiculopathies fulfilled the inclusion criteria based on having fibrillations and positive sharp waves and not more subjective neuropathic changes. Radiculopathy levels were defined by the group of muscles that were abnormal according to generally accepted anatomic innervation levels.[6] The minimum number of root levels that explained all abnormal muscles defined the radiculopathy levels. When the precise root level could not be accurately delineated, or when 2 levels were equally as likely to explain the pattern of muscle findings, the subject was classified by terminology that indicated these 2 levels; i.e., L4–5. This does not refer to herniation of an L4–5 nucleus pulposis. In this sample of patients with electrodiagnostically confirmed lumbosacral radiculopathies, 9 involved only the L3–4 levels, 12 the L4–5 levels, 29 the L5 level, 21 the L5–S1 levels, 19 at the S1 level, 4 involved multiple (>2) levels, and 8 were indeterminate, with only the paraspinal muscles demonstrating abnormalities. In 85% of the cases, patients reported leg pain, and 72% reported low-back pain. Based on a pain drawing, 50% of the sample also described suffering from radicular patterns of pain.

TABLE 1
Individual muscle identifications of patients with lumbosacral radiculopathies

Muscle	Percentage Studied (%)	Neuropathic (%)	Spontaneous Activity (%)
Paraspinals	88	61	60
Posterior tibialis	92	61	42
Anterior tibialis	100	53	38
Medial gastrocnemius	99	37	33
Tensor fasciae latae	99	35	25
Lateral gastrocnemius	99	27	21
Short-head biceps femoris	98	23	15
Vastus medialis	99	15	7
Vastus lateralis	98	13	7
Rectus femoris	98	12	6
Adductor longus	100	14	9

The muscle successfully identified a radiculopathy if that muscle was abnormal for the patient. For the neuropathic findings column, patients with a radiculopathy were identified if the nonparaspinal muscle showed positive waves, fibrillations, increased polyphasic potentials, neuropathic recruitment, increased insertional activity, complex repetitive discharges, or large-amplitude, long-duration motor-unit action potentials. In the spontaneous activity column, the muscle had to demonstrate fibrillations or positive sharp waves. For paraspinal muscles, the neuropathic category included fibrillations, increased insertional activity, positive waves, and complex repetitive discharges. The "percentage studied" column refers to the percentage of patients who had this muscle examined.

Individual Muscle Identifications

Table 1 shows identification rates for each individual muscle. The paraspinal muscles identified the majority (61%) of radiculopathies. For each muscle, when using the neuropathic definition, the identification rates were 1–19% higher than when only spontaneous activity defined an abnormal muscle. Most muscles were studied in >98% of patients.

The medial gastrocnemius demonstrated a significantly higher identification rate for all radiculopathies ($P < 0.001$ by Pearson's correlation) than the lateral gastrocnemius. Table 1 shows that when using both neuropathic findings and spontaneous activity categories, the medial gastrocnemius identified approximately 10% more of the sample than the lateral gastrocnemius. Furthermore, when analyzing the subset of patients with L5–S1 and S1 radiculopathies, the medial gastrocnemius demonstrated significantly ($P < 0.04$) higher identification than the lateral gastrocnemius.

None of the L3 or L4 innervated muscles (VMED, RFEM, VLAT, or ADD) demonstrated significantly different identifications for the entire group (Table 1). Likewise, for the subset of L3–4 radiculopathies and L4–5 radiculopathies, these muscles had no significant differences in their identification rates.

Muscle Screen Identifications

Muscle screen identification rates for electrodiagnostically confirmed lumbosacral radiculopathies are shown in Tables 2-6. These muscle screens consisted of combinations of muscles representing all root levels based on their innervations.

Table 2 shows identification rates for four-muscle screens. When the more inclusive neuropathic definition was used, identification rates were substantially higher than when only spontaneous activity was considered. Like-

TABLE 2
Four-muscle screen identifications of patients with lumbosacral radiculopathies

Screen	Neuropathic (%)	Spontaneous Activity (%)
Four muscles without paraspinals		
ATIB, PTIB, MGAS, RFEM	85	75
VMED, TFL, LGAS, PTIB	75	58
VLAT, SHBF, LGAS, ADD	52	35
ADD, TFL, MGAS, PTIB	80	67
Four muscles with paraspinals		
ATIB, PTIB, MGAS, PSM	97	90
VMED, LGAS, PTIB, PSM	91	81
VLAT, TFL, LGAS, PSM	88	77
ADD, MGAS, PTIB, PSM	94	86

ATIB, tibialis anterior; PTIB, tibialis posterior; MGAS, medial gastrocnemius; RFEM, rectus femoris; VMED, vastus medialis; TFL, tensor fasciae latae; LGAS, lateral gastrocnemius; VLAT, vastus lateralis; SHBF, short head of the biceps femoris; ADD, adductor longus; PSM, lumbosacral paraspinal muscles.

The screen identified the patient if any muscle in the screen was abnormal for that patient. The muscle either demonstrated neuropathic findings or spontaneous activity. Neuropathic findings for nonparaspinal muscles included positive waves, fibrillations, increased polyphasic potentials, neuropathic recruitment, increased insertional activity, complex repetitive discharges, or large-amplitude, long-duration motor-unit action potentials. Spontaneous activity referred only to fibrillations or positive sharp waves. For paraspinal muscles the neuropathic category included fibrillations, increased insertional activity, positive waves, or complex repetitive discharges.

TABLE 3
Five-muscle screen identifications of patients with lumbosacral radiculopathies

Screen	Neuropathic (%)	Spontaneous Activity (%)
Five muscles without paraspinals		
ATIB, PTIB, MGAS, RFEM, SHBF	88	77
VMED, TFL, LGAS, PTIB, ADD	76	59
VLAT, SHBF, LGAS, ADD, TFL	68	50
ADD, TFL, MGAS, PTIB, ATIB	86	78
Five muscles with paraspinals		
ATIB, PTIB, MGAS, PSM, VMED	98	91
VMED, LGAS, PTIB, PSM, SHBF	97	84
VLAT, TFL, LGAS, PSM, ATIB	97	86
ADD, MGAS, PTIB, PSM, VLAT	94	86

Abbreviations are defined in Table 2.

wise, when muscle screens did not include paraspinal muscles, identification rates were lower. For four-muscle screens with paraspinal muscles, identification rates ranged from 88% to 97% with neuropathic findings and from 77% to 90% when only spontaneous activity was considered.

As shown in Table 3, five-muscle screens yielded identification rates of 94–98% when paraspinal muscles were included. Without the paraspinal muscles, 68–88% were identified. These findings indicate that five-muscle screens that include paraspinal muscles and examine neuropathic findings provide excellent detection of an electrodiagnostically confirmable lumbosacral radiculopathy.

Six-muscle screen identifications are shown in Table 4. These screens identified 98–100% of the sample. By adding one muscle over a five-muscle screen, slightly higher identification rates were achieved.

Seven-muscle screen identifications are shown in Table 5. Identification rates with paraspinal muscles were 99–100%. When the paraspinal muscles were not included, identifications ranged from 84% to 90%.

Eight-muscle screens without paraspinals are shown in Table 6. The purpose of this analysis was to determine, for those patients in whom the lumbar paraspinals were unreliable, how many lower-limb muscles should be studied to reach a maximal level of identification. These screens correctly detected 87–90% of confirmable radiculopathies. Nine- and 10-muscle screens did not yield further improvements in identification rates when the paraspinals were not included. This finding suggests that whenever the paraspinal muscles were unreliable for inclusion in the screen (for example, in patients with previous lumbosacral spine surgery), 8 lower-limb muscles should be studied.

DISCUSSION

Determining how many and which muscles to study to identify successfully a lumbosacral radiculopathy (when one can be electrodiagnostically confirmed) is an important clinical issue. The results of this study indicate that a six-muscle needle EMG screen, including paraspinal muscles, has a high detection rate for a lumbosacral radiculopathy that can be confirmed by EMG. Studies involving seven or more muscles provide

TABLE 4
Six-muscle screen identifications of patients with lumbosacral radiculopathies

Screen	Neuropathic (%)	Spontaneous Activity (%)
Six muscles without paraspinals		
ATIB, PTIB, MGAS, RFEM, SHBF, LGAS	89	78
VMED, TFL, LGAS, PTIB, ADD, MGAS	83	70
VLAT, SHBF, LGAS, ADD, TFL, PTIB	79	62
ADD, TFL, MGAS, PTIB, ATIB, LGAS	88	79
Six muscles with paraspinals		
ATIB, PTIB, MGAS, PSM, VMED, TFL	99	93
VMED, LGAS, PTIB, PSM, SHBF, MGAS	99	87
VLAT, TFL, LGAS, PSM, ATIB, SHBF	98	87
ADD, MGAS, PTIB, PSM, VLAT, SHBF	99	89
VMED, ATIB, PTIB, PSM, SHBF, MGAS	100	92
VMED, TFL, LGAS, PSM, ATIB, PTIB	99	91
ADD, MGAS, PTIB, PSM, ATIB, SHBF	100	93

Abbreviations are defined in Table 2.

TABLE 5
Seven-muscle screen identifications of patients with lumbosacral radiculopathies

Screen	Neuropathic (%)	Spontaneous Activity (%)
Seven muscles without paraspinals		
ATIB, PTIB, MGAS, RFEM, SHBF, LGAS, ADD	90	79
VMED, TFL, LGAS, PTIB, ADD, MGAS, RFEM	84	71
VLAT, SHBF, LGAS, ADD, TFL, PTIB, MGAS	85	71
ADD, TFL, MGAS, PTIB, ATIB, LGAS, VLAT	88	79
Seven muscles with paraspinals		
ATIB, PTIB, MGAS, PSM, VMED, TFL, LGAS	100	94
VMED, LGAS, PTIB, PSM, SHBF, MGAS, TFL	99	90
VLAT, TFL, LGAS, PSM, ATIB, SHBF, ADD	99	89
ADD, MGAS, PTIB, PSM, VLAT, SHBF, RFEM	99	89

Abbreviations are defined in Table 2.

no better identification than six muscles. If the paraspinal muscles cannot be reliably studied[23–27], eight distal muscles are necessary to identify up to 90% of lumbosacral radiculopathies. It is important to note that if an electrodiagnostician can only study four muscles (including paraspinals) because of limited patient tolerance, only a few confirmable lumbosacral radiculopathies will be missed; identification based on four-muscle screens (including paraspinals) ranged from 88% to 97%.

The use of an EMG screening evaluation is to alert the examiner to the possibility of a radiculopathy. If one of the screening muscles is abnormal, then the examiner must expand and tailor the study to exclude other disorders such as polyneuropathy, motor neuron disease, or entrapment neuropathy. Likewise, additional muscles are usually needed to identify the actual electrodiagnostic root level.

A broader group of neuropathic findings were analyzed and showed higher identification rates than with spontaneous activity alone. Hoover et al.[28] demonstrated the importance of recognizing polyphasic potentials when evaluating for a lumbosacral radiculopathy. Because of the screening nature of EMG assessed in this study, a broader, more inclusive neuropathic group of findings were analyzed. Although more subjective than spontaneous activity and more prone to false positivity, these neuropathic findings yielded higher identifications. Therefore, assessment of more subtle findings of motor axonal loss such as polyphasic potentials and reduced recruitment is recommended in the context of a screening EMG examination. If such findings are recognized, the study must be expanded to determine whether these reflect a radiculopathy (if found in combination with other myotomal abnormalities), a mononeuropathy (if limited to a single nerve distribution), a generalized condition (if found in other limbs), or whether they are of no clinical significance. Only an experienced electrodiagnostician can render this judgment.

The muscle screen provides the electrodiagnostician with a limited number of muscles that, if normal, can reliably exclude an electrodiagnostically confirmable radiculopathy. The patient might still have a radiculopathy by clinical or imaging criteria, but this diagnosis cannot be confirmed by needle EMG, and a more extensive study is not indicated.

This article does not address the concept of EMG sensitivity relative to a diagnostic standard such as clinical or imaging findings. Rather, this study dealt with the issue of how many muscles must be studied by needle EMG to detect when a lumbosacral radiculopathy can be electrodiagnostically confirmed. This is a fundamentally different concept than sensitivity. Some radiculopathies do not demonstrate EMG findings and cannot be detected no matter how many muscles are tested. The mini-

TABLE 6
Eight-muscle screen identifications of patients with lumbosacral radiculopathies

Screen	Neuropathic (%)	Spontaneous Activity (%)
Eight muscles without paraspinals		
ATIB, PTIB, MGAS, RFEM, SHBF, LGAS, ADD, TFL	90	81
VMED, TFL, LGAS, PTIB, ADD, MGAS, RFEM, ATIB	89	80
VLAT, SHBF, LGAS, ADD, TFL, PTIB, MGAS, RFEM	87	73
ADD, TFL, MGAS, PTIB, ATIB, LGAS, VLAT, SHBF	89	80
ATIB, PTIB, MGAS, SHBF, VMED, TFL, LGAS, ADD	90	81
VMED, LGAS, PTIB, RFEM, SHBF, MGAS, TFL, ATIB	90	80

Abbreviations are defined in Table 2.

mum number of muscles necessary to recognize a radiculopathy that can be delineated by EMG is an important clinical issue. The results of this study suggest that selected six-muscle screens are optimal for this detection purpose.

Screens involving more than six muscles (including paraspinals) are not indicated unless there is a clinical suspicion that another neurologic condition might be present. Nonetheless, these muscle screens should not substitute for a clinical evaluation and differential diagnosis formulation by the electrodiagnostic consultant. Rather, information from these screens allows the electrodiagnostician to tailor the EMG evaluation and make informed clinical decisions regarding the probability of missing a radiculopathy when a given set of muscles is studied. Performing a focused history and physical examination is essential, and these screens should not supplant such clinical assessment.

Paraspinal muscles demonstrate high identification rates for lumbosacral radiculopathies by themselves and substantially augment the identification rates for all screens.[15, 16] Although they are technically more challenging to study[22], their inclusion is recommended in all screening examinations whenever possible. If paraspinal muscles cannot be studied, then eight distal muscles should be examined. Our inclusion criteria ensured that patients who had previous lumbosacral spine surgery were not included based on paraspinal findings, but rather demonstrated abnormalities in other nonparaspinal muscles. For this reason, the results of this study can be generalized to patients with previous spine surgery but require the examiner to study eight lower-limb muscles to confidently identify about 90% of patients with radiculopathy.

Two important limitations of this study merit mention. First, 11 standard muscles were chosen based on the authors' previous study because they are easily identified and because they are innervated by major lower-limb peripheral nerves. Other muscles may be equally as effective at identification. Second, lumbosacral spinal assessment with magnetic resonance imaging was not available for the entire sample. Magnetic resonance imaging is an important assessment tool that can provide valuable information about structural causes of radiculopathy.

The strength of this study follows from its multicenter, prospective nature and the use of a standard screen. This ensured the generalizability of our results to a typical referral population seen at most electrodiagnostic laboratories. It is reassuring that five- and six-muscle screens yielded such high identification rates. Had it not been so, we would be concerned that a ceiling effect had taken place. For instance, if 9- or 10-muscle screens were necessary to provide high identification rates, it would be apparent that our standard screen should have included >11 muscles. This, however, was not the case, and six muscles identified almost all radiculopathies, further supporting the validity of our findings. Presumably, the 11-muscle standard screen identified all lumbosacral radiculopathies that could be identified with standard EMG, and it is unlikely that studying more muscles would have improved identification.

Incorporating these findings into the overall clinical assessment and management of patients with low-back and lower-limb symptoms and determining their usefulness in delineating and predicting treatment outcomes are important areas of future research.

CONCLUSIONS

Based on the findings of this study, six lower-limb muscles, including paraspinal muscles, consistently identify almost all electrodiagnostically confirmable lumbosacral radiculopathies. Studying additional muscles does not increase identification. When paraspinal muscles are not reliable to study, eight distal muscles are necessary to achieve acceptable identification.

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